

- CARLOS ERNESTO RAMÍREZ OVALLE, *A Concrete Symplectic Semantics for (Intuitionistic) Linear Logic with Labelled Subexponentials*.

Department of Mathematics, Pontificia Universidad Javeriana–Cali, Colombia.

E-mail: carlosovalle@javerianacali.edu.co.

We propose a concrete symplectic semantics for (intuitionistic) linear logic with *labelled subexponentials* inside the category **LagRel** of finite-dimensional symplectic vector spaces and *Lagrangian relations*. Beyond its interest for logic and theoretical computer science, our approach highlights structural facts that are purely categorical and symplectic: we exhibit a $*$ -autonomous structure on **LagRel** and use it to interpret the multiplicative-additive fragment together with label-sensitive structural rules. On the logical side we prove *soundness* (every derivable sequent is interpreted by a Lagrangian relation) and a form of *relative completeness/fullness* (every Lagrangian relation of the appropriate type is denoted by a cut-free proof once labels and context formation are fixed). On the linear-algebraic side we provide explicit *matrix characterisations* for the connectives and rules, yielding a hands-on calculus; short worked examples in low dimension (e.g. 2×2 blocks) illustrate how label constraints are reflected at the level of Lagrangian subspaces. Altogether, the results deliver a mathematically transparent bridge between cut-elimination and composition of Lagrangian correspondences, and they furnish a concrete $*$ -autonomous model of labelled subexponentials grounded in classical symplectic geometry.